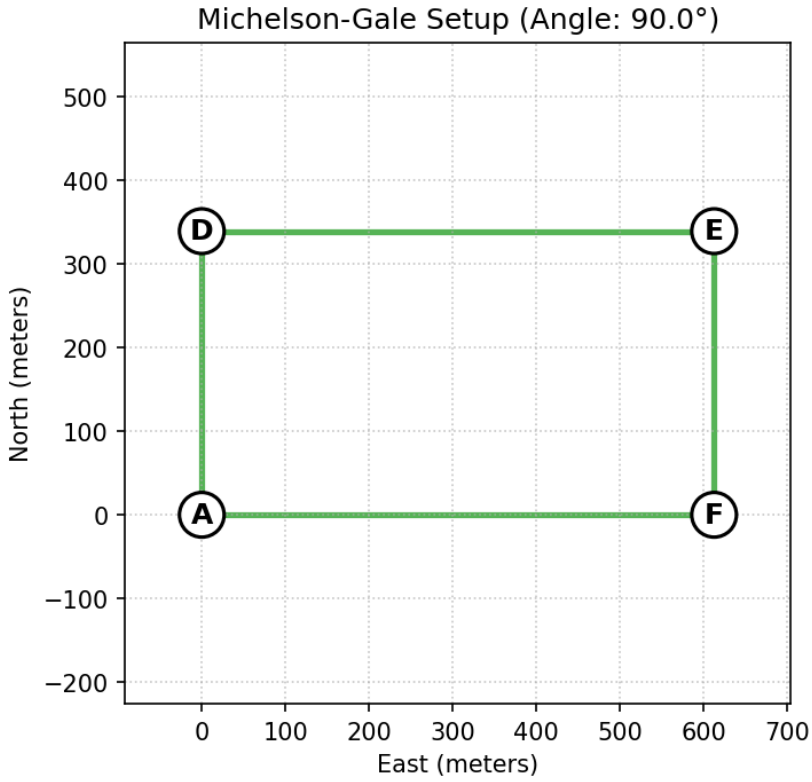
3D ECI Interferometer Simulation Report

SCENARIO 1a: Michelson-Gale: No Earth rotation + No Drift Vector




```
Location:       41.76666666666667 deg N
Loop Shape:     A-D: 339.2424m, A-F: 612.648m @ Angle: 90.0 deg
Wavelength:     570e-9 m
Kinematics:     Earth Rot: OFF | Drift Vector: 0.0 m/s @ 0.0 deg

--- Michelson-Gale (0 deg Orientation) ---
Time Diff:      0.0000000000 ns
Distance Diff:  0.0000000000 nm
Fringe Shift:   0.0000000000 fringes
```

SCENARIO 1b (as expected): Michelson-Gale: Earth rotation + No Drift Vector

```
Location:       41.76666666666667 deg N
Loop Shape:     A-D: 339.2424m, A-F: 612.648m @ Angle: 90.0 deg
Wavelength:     570e-9 m
Kinematics:     Earth Rot: ON | Drift Vector: 0.0 m/s @ 0.0 deg

--- Michelson-Gale (0 deg Orientation) ---
Time Diff:      0.0000004493 ns
Distance Diff:  134.6953190521 nm
Fringe Shift:   0.2363075773 fringes
```

SCENARIO 1c: Michelson-Gale: Earth rotation + Drift Vector

```
Location:       41.76666666666667 deg N
Loop Shape:     A-D: 339.2424m, A-F: 612.648m @ Angle: 90.0 deg
```

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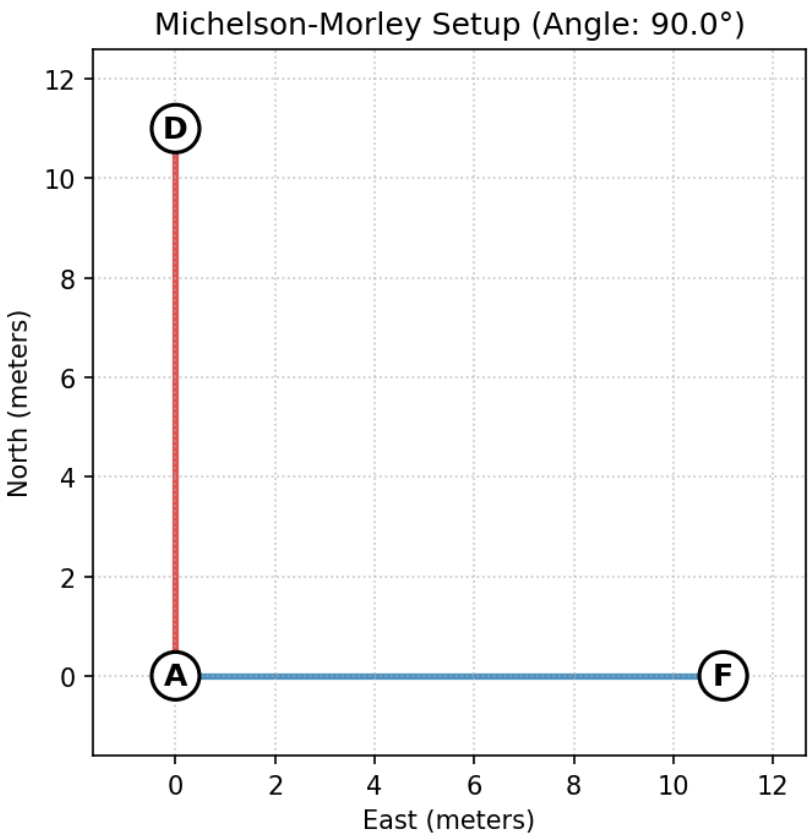
```
Wavelength:     570e-9 m
Kinematics:     Earth Rot: ON | Drift Vector: 30000.0 m/s @ 0.0 deg

--- Michelson-Gale (0 deg Orientation) ---
Time Diff:      0.0000004493 ns
Distance Diff:  134.6953211299 nm
Fringe Shift:   0.2363075809 fringes
```

SCENARIO 2a (as expected): Michelson-Morley: Earth rotation + No Drift Vector




```
Location:       41.5 deg N
Loop Shape:     A-D: 11m, A-F: 11m @ Angle: 90.0 deg
Wavelength:     590e-09 m
Kinematics:     Earth Rot: ON | Drift Vector: 0.0 m/s @ 0.0 deg

--- Michelson-Morley (0 deg Orientation) ---
Time Diff:      0.0000000000 ns
Distance Diff:  0.0148179157 nm
Fringe Shift:   0.0000251151 fringes
```

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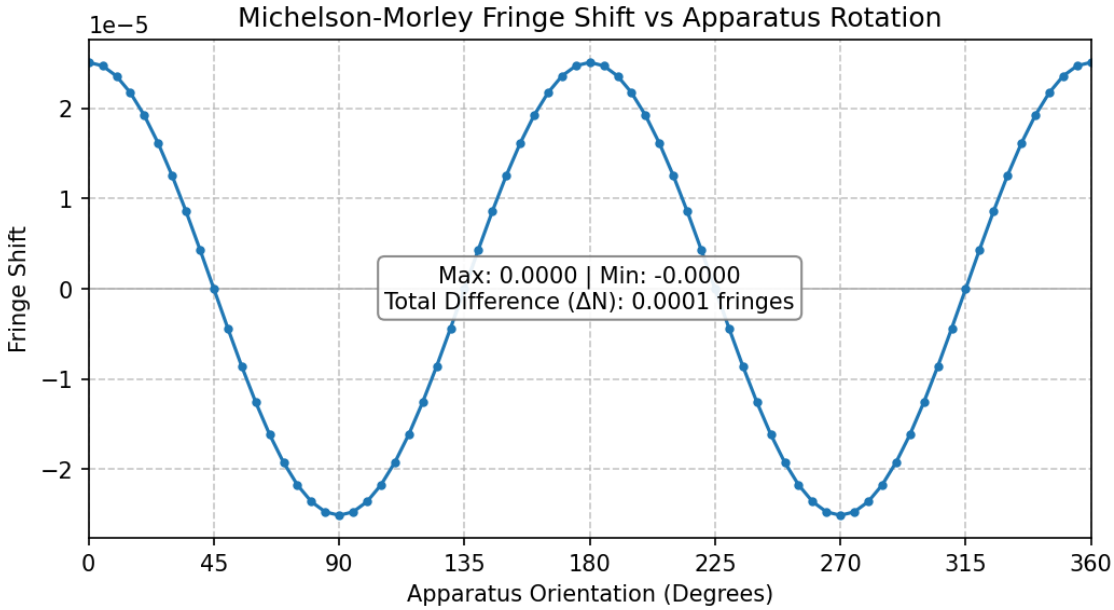
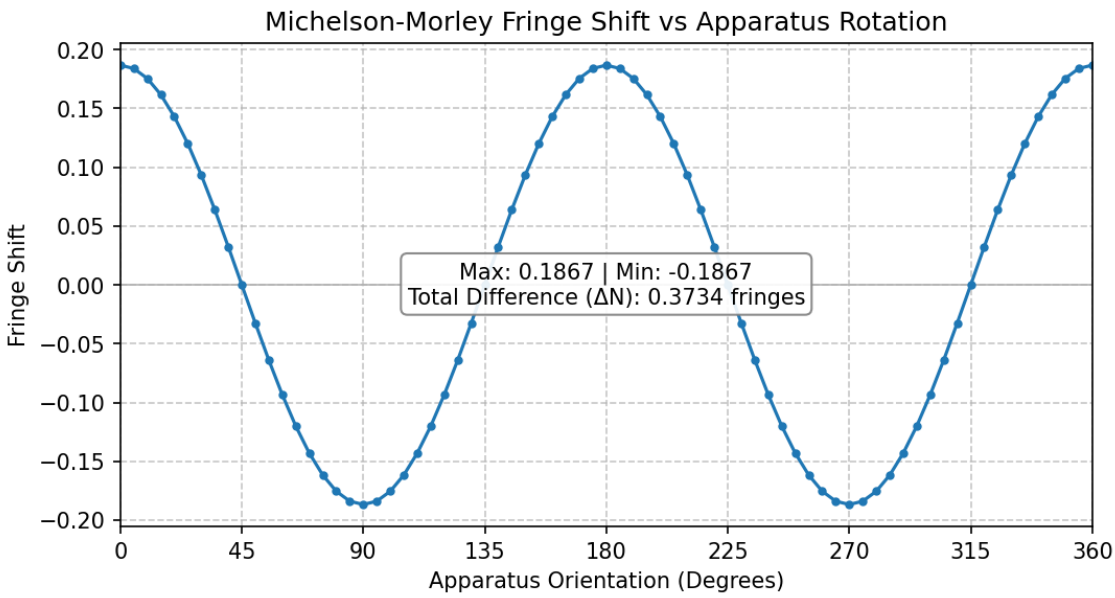


SCENARIO 2b: Michelson-Morley: No Earth rotation + Drift Vector

```
Location:       41.5 deg N
Loop Shape:     A-D: 11m, A-F: 11m @ Angle: 90.0 deg
Wavelength:     590e-09 m
Kinematics:     Earth Rot: OFF | Drift Vector: 30000.0 m/s @ 0.0 deg

--- Michelson-Morley (0 deg Orientation) ---
Time Diff:      0.0000003674 ns
Distance Diff:  110.1523569281 nm
Fringe Shift:   0.1866989100 fringes
```



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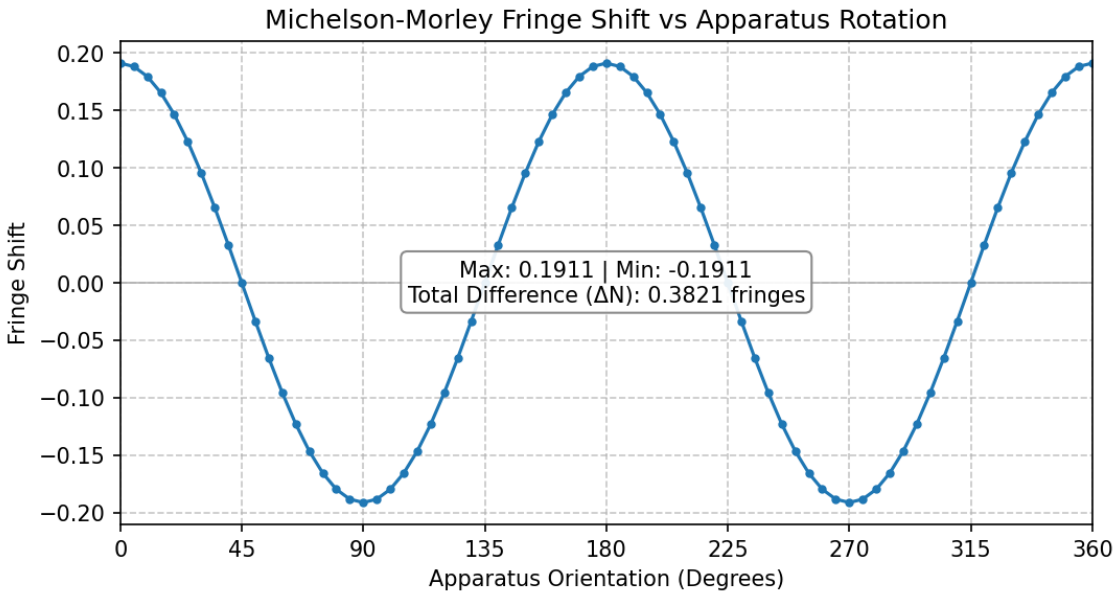
SCENARIO 2c: Michelson-Morley: Earth rotation + Drift Vector

```
Location:       41.5 deg N
Loop Shape:     A-D: 11m, A-F: 11m @ Angle: 90.0 deg
Wavelength:     590e-09 m
Kinematics:     Earth Rot: ON | Drift Vector: 30000.0 m/s @ 0.0 deg

--- Michelson-Morley (0 deg Orientation) ---
Time Diff:      0.0000003760 ns
Distance Diff:  112.7223467806 nm
Fringe Shift:   0.1910548251 fringes
```



SCENARIO 3a: Einstein telescope; Earth doesn't rotate

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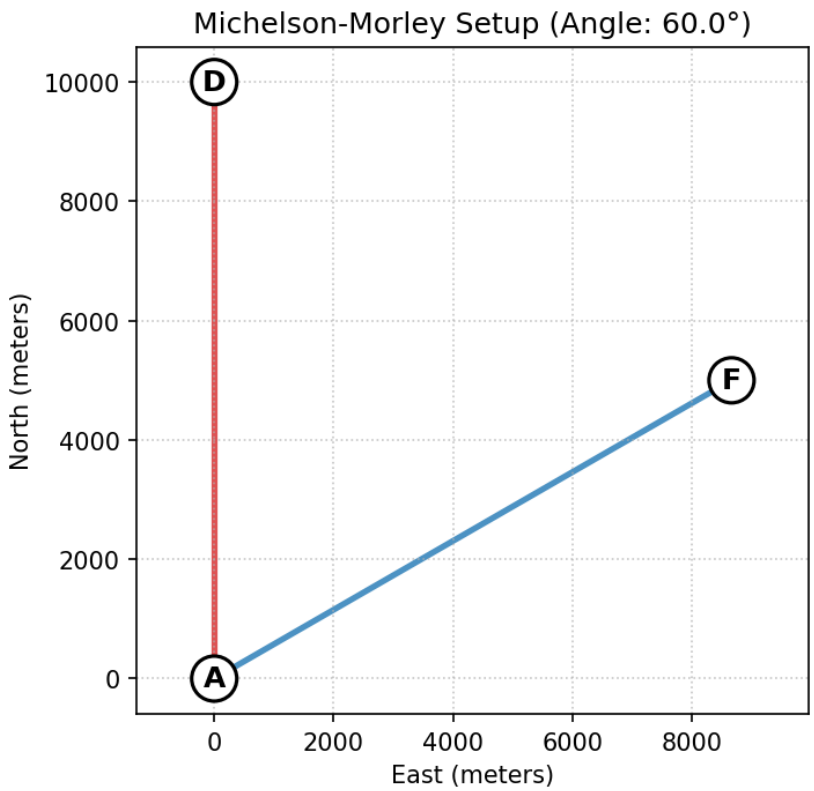


```
Location:       50.7542 deg N
Loop Shape:     A-D: 10000.0m, A-F: 10000.0m @ Angle: 60.0 deg
Wavelength:     1.55e-06 m
Kinematics:     Earth Rot: OFF | Drift Vector: 0.0 m/s @ 0.0 deg

--- Michelson-Morley (0 deg Orientation) ---
Time Diff:      -0.0000000000 ns
Distance Diff:  -0.0000000000 nm
Fringe Shift:   -0.0000000000 fringes
```

SCENARIO 3b (as expected): Einstein telescope; Earth rotates

```
Location:       50.7542 deg N
Loop Shape:     A-D: 10000.0m, A-F: 10000.0m @ Angle: 60.0 deg
Wavelength:     1.55e-06 m
Kinematics:     Earth Rot: ON | Drift Vector: 0.0 m/s @ 0.0 deg

--- Michelson-Morley (0 deg Orientation) ---
Time Diff:      0.0000000241 ns
Distance Diff:  7.2319295728 nm
Fringe Shift:   0.0046657610 fringes
```